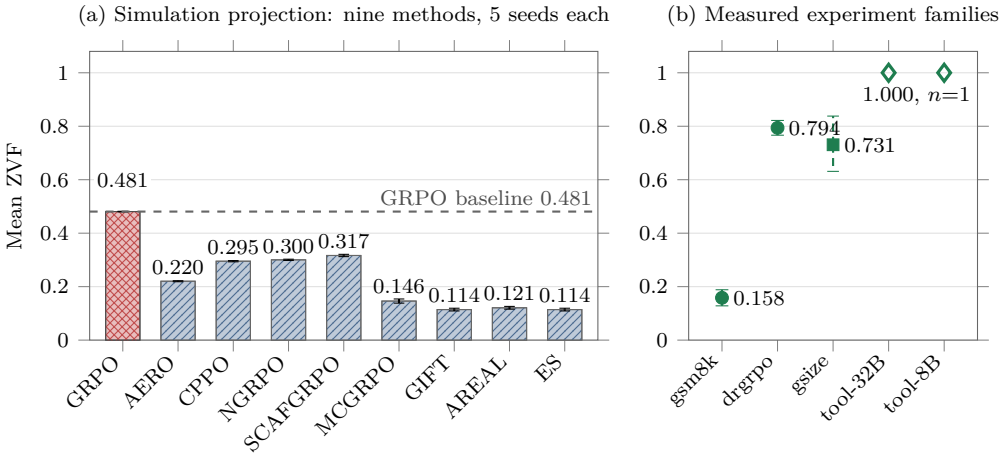
(a) Simulation projection: nine methods, 5 seeds each
(b) Measured experiment families
Mean ZVF
1
0.8
0.6
0.4
0.2
0
0.481
GRPO baseline 0.481
0.220
0.295
0.300
0.317
0.146
0.114
0.121
0.114
GRPO
AERO
CPPO
NGRPO
SCAFGRPO
MCGRPO
GIFT
AREAL
ES
1.000, n=1
0.794
0.731
0.158
gsm8k
drgrpo
gsize
tool-32B
tool-8B